CYBERSECURITY / PRACTICAL SERIES

WEEK 02

RECONNAISSANCE
AND LOCAL NETWORK DISCOVERY

A practical report on domain footprinting and Zenmap host discovery

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Practical activities	W2-PM1 - Kali Linux footprinting | W2-PM5 - Zenmap scanning
Assessment date	11 November 2026
Programme / batch	B084
Authorised environments	networkwalks.com (written permission stated) and my own local LAN

SUBMITTED AS PART OF THE WEEK 2 INTERNSHIP PRACTICAL

01 / PURPOSE AND SCOPE

Report overview

This report records two practical exercises completed during Week 2 of my Cybersecurity & Ethical Hacking internship. The first exercise focused on collecting publicly visible information about networkwalks.com. The second used Zenmap to identify devices responding on my own home/local network.

MAIN OUTCOME
The footprinting exercise produced a profile of the domain's registration, web technologies, HTTP response and DNS records. The local scan returned four responding hosts on the 192.168.8.0/24 subnet.

What I set out to do

01 Collect different categories of public domain information using six Kali Linux utilities.

02 Identify the IP configuration of my local network and use Zenmap Ping Scan to discover responding devices.

03 Record the findings, explain why they matter to a defender, and suggest reasonable security checks.

Authorisation and boundaries

The domain footprinting activity was conducted within the written permission stated for networkwalks.com. The Zenmap activity was restricted to my own local LAN. The work described here covers information gathering and host discovery only. I did not attempt to exploit the website, bypass security controls, access device contents or validate vulnerabilities.

How to read this report

For each practical, I describe the activity, note the output recorded during the exercise and explain what the observation can tell a security practitioner. The evidence register at the end identifies the screenshots that should accompany the final submission.

IMPORTANT LIMITATION
The findings reflect what the tools returned at the time of testing. They should not be treated as proof that a system is secure, insecure, compromised or vulnerable.

02 / WORKING ENVIRONMENT

Lab setup and tools

I used two environments because the practicals had different purposes. Kali Linux was used for domain footprinting, while Zenmap was run from a Windows PC for local network discovery.

Environment / utility	Role in the exercise
Kali Linux	Operating system used to run the domain-footprinting utilities.
WHOIS	Looked up public domain registration details, including registrar, dates and name-server information.
WhatWeb	Identified technologies and components that the website disclosed.
nslookup	Queried DNS to resolve the domain name to an IP address.
curl -I	Requested the HTTP response headers without downloading the full webpage.
Wafw00f	Checked for an identifiable Web Application Firewall.
DNSRecon	Collected DNS information, including SOA, MX, TXT and service records.
Windows Command Prompt	Used ipconfig to view the local PC network configuration.
Zenmap (Nmap GUI)	Used the Ping Scan profile to discover responding hosts and view the topology.

Process followed

For the domain exercise, I ran each of the six utilities separately and recorded the information relevant to its purpose. For the LAN exercise, I checked the PC's addressing details first, selected the local subnet in Zenmap, ran Ping Scan and then reviewed the Topology view with the legend enabled.

> **COMMAND RECORD**
> The report identifies the tools and the one explicit option recorded in the practical brief (curl -I). Where a screenshot is required, retain the original command line visible in the terminal rather than adding a command that was not actually run.

03 / W2-PM1

Practical A: Domain footprinting

I examined networkwalks.com with six tools. Instead of treating the outputs as six unrelated results, I grouped the findings into domain and DNS information, followed by website technology and response information.

A. Domain registration and DNS picture

WHOIS - registration information

The WHOIS result showed GoDaddy as the registrar, a creation date of 6 November 2019 and name servers associated with HostGator. This gave me an initial picture of the domain's registration and the providers reflected in its public records. These details are useful when documenting ownership and hosting arrangements, but they do not establish that the domain or its hosting account is exposed to attack.

nslookup - domain resolution

The nslookup result resolved networkwalks.com to 192.232.216.135. This is the IP address returned for the domain during my query. It identifies a network endpoint associated with the website at that moment; it should not automatically be assumed to represent every server or service used by the organisation.

DNSRecon - published DNS records

DNSRecon returned SOA information associated with HostGator, mail-exchange (MX) records, TXT records and an SRV record referring to cpanelmaildiscovery.cpanel.net. The results helped me understand how DNS can reveal details about domain administration, email handling and related services. Each record would need to be checked in context before deciding whether it creates a security concern.

> **WHAT I LEARNED**
>
> WHOIS and DNS tools provide different parts of the same picture. Registration information describes the domain's administrative background, while DNS records show how selected services are named and directed.

03 / W2-PM1 - CONTINUED

Practical A: Website technology and response

B. Web-facing information

WhatWeb - technology identification

WhatWeb identified Apache and reported the IP address 192.232.216.135. The output also showed an HTTP-to-HTTPS redirect, cookies, WordPress 7.1.4, WordPress Download Manager 3.3.76 and jQuery 3.7.1. This information can support a software inventory and help an administrator check whether the reported components are current. A detected version is only a lead for review; it is not proof of a known or exploitable weakness.

curl - response headers

Using curl with the -I option, I received an HTTP/2 200 response and saw the Server: Apache header. The response also included headers such as X-Frame-Options and Content-Type. This helped me see how the server describes its response to a request. The status code indicates that the request was handled successfully; it does not measure the security of the application.

Wafw00f - firewall identification

Wafw00f identified ModSecurity (SpiderLabs) as the Web Application Firewall technology associated with the website. This is useful context when documenting the site's visible security controls. However, identifying a WAF does not show whether its rules are effective, correctly maintained or sufficient for the application.

Summary of the footprinting results

Information area	Recorded observation	Reason for keeping the result
Domain administration	GoDaddy registrar; creation date 6 November 2019; HostGator name servers.	Documents the public registration and name-server details.
DNS resolution	networkwalks.com resolved to 192.232.216.135.	Records the endpoint returned by the DNS query.
Web stack	Apache; WordPress 7.1.4; Download Manager 3.3.76; jQuery 3.7.1.	Supports an inventory and update review.
HTTP response	HTTP/2 200; Apache header; X-Frame-Options and Content-Type observed.	Provides context for response and header configuration.
Security control	ModSecurity (SpiderLabs) identified.	Records the WAF technology detected by the tool.
DNS services	SOA, MX, TXT and SRV information observed.	Helps document published domain, mail and service records.

04 / W2-PM5

Practical B: My local network

For this exercise, I used my Windows PC to identify the local network configuration before starting Zenmap. The ipconfig output indicated the subnet 192.168.8.0/24. I entered this subnet in Zenmap and selected the Ping Scan profile to check which hosts responded.

SCAN RESULT
Zenmap reported four live hosts on the subnet during the scan. This count includes my local PC.

Responding hosts recorded

No.	IP address	Device description	MAC address
01	192.168.8.1	Router / default gateway	22:2F:48:F4:D8:D3
02	192.168.8.36	Mobile device (identity withheld)	92:6F:2E:0B:36:03
03	192.168.8.60	My local PC	Not shown in output
04	192.168.8.70	Unknown device	DA:22:54:8A:49:DE

Topology view

After the scan, I opened Zenmap's Topology section, enabled the legend and saved the network topology as a PDF named Zenmap_Topology.pdf. The topology view gives a visual summary of the responding hosts and shows the router as the central network point in the map.

Interpreting this scan

The one entry marked as unknown (192.168.8.70) was not identified from the information recorded in this exercise. It should remain labelled as unknown rather than being assigned to a person or device type without further confirmation. The mobile device identity is withheld to protect privacy.

SCAN LIMITATION

A Ping Scan reports hosts that respond to the scan at that time. A device that is switched off or configured not to respond may not appear. The results are a snapshot of network visibility, not a complete inventory.

05 / RISK REVIEW

What the findings may mean

The exercise identified information that a system owner may want to review. The table below assesses the possible significance of each observation. The ratings are preliminary and should not be read as confirmed vulnerabilities.

No.	Observation	Evidence	Possible implication	Priority
01	Technology and plugin versions were visible	WhatWeb reported WordPress 7.1.4, Download Manager 3.3.76 and jQuery 3.7.1.	The owner should confirm that the components are supported and patched.	Medium
02	The domain resolved to a public IP address	nslookup returned 192.232.216.135.	This identifies a public endpoint, but an address alone is not a weakness.	Low
03	HTTP response details were available	curl returned server and response-header information.	The owner can compare the headers with its configuration standard.	Low
04	WAF technology could be identified	Wafw00f reported ModSecurity (SpiderLabs).	This describes a visible control but does not show whether it is effective.	Low
05	DNS records described domain services	DNSRecon returned SOA, MX, TXT and SRV records.	The organisation should verify that published records are still required.	Medium
06	Four hosts responded on the LAN	Zenmap identified four live hosts, including one device not identified in the notes.	The network owner should compare the unknown entry with an approved device list.	Medium

How I assigned the priorities

I used Medium where the observation suggests a practical follow-up check, such as confirming software updates or identifying an unknown device. I used Low where the result mainly describes information that is normally visible during routine network communication. These are prioritisation labels for this report, not the result of a formal vulnerability-scoring exercise.

OVERALL ASSESSMENT

The activities were limited to reconnaissance and host discovery. No exploitation, authentication testing or vulnerability validation was carried out. An exposed version string or an unknown live host is a reason to investigate, not enough evidence to conclude that a system has been compromised.

06 / PRACTICAL IMPROVEMENTS

Recommended follow-up actions

The recommendations below focus on checks that can be completed by the relevant website or network owner. They are based on the observations in this report rather than on an assumption that a vulnerability has already been found.

For the website and domain owner

01 Review the public technology footprint. Check which server, CMS and plugin details are exposed and decide whether each disclosure is necessary for normal operation.

02 Maintain a patching routine. Confirm the current supported versions of WordPress, Download Manager, jQuery and the web server, then apply updates through the normal change process.

03 Review HTTP security settings. Compare the observed response headers with the organisation's security baseline and correct any gaps identified by that review.

04 Check DNS records. Confirm that SOA, MX, TXT and SRV records are accurate, maintained and required for active services.

05 Maintain the WAF. Review ModSecurity configuration, update its rules and inspect logs as part of normal security monitoring.

For the local network owner

06 Verify the unknown device. Compare the one unidentified entry (192.168.8.70) with devices used by the household or network owner before deciding whether any action is needed.

07 Keep an up-to-date device list. Record approved devices, their owners or purpose, and relevant network details so future scans can be compared with a known baseline.

08 Repeat discovery responsibly. Conduct network discovery periodically and only on networks where permission has been given.

My learning reflection

This practical helped me understand that reconnaissance is not a single tool or command. Each utility reveals a different piece of information, and the value comes from interpreting the results together. I also learned to be careful with conclusions: a tool can identify a technology or a responding device, but that observation alone does not prove that something is vulnerable.

Conclusion

I completed the Week 2 footprinting and network-scanning activities using six Kali Linux tools and Zenmap. The domain exercise gave me information about registration, DNS, website technologies, HTTP headers and WAF identification. The LAN exercise showed four responding hosts on my 192.168.8.0/24 network and allowed me to produce a topology view. Overall, the work strengthened my understanding of how information is collected and documented during an authorised security assessment. It also reinforced the need to keep testing within the agreed scope and to support conclusions with evidence.

APPENDIX A / SCREENSHOTS

Evidence register

Insert the original screenshots in the spaces below. Each image should show enough of the tool output to support the observation recorded in the report. Keep text readable and remove personal details that are not needed for assessment.

FIGURE 01 | WHOIS - domain registration

WHOIS terminal output showing the registrar, domain creation date and name-server information

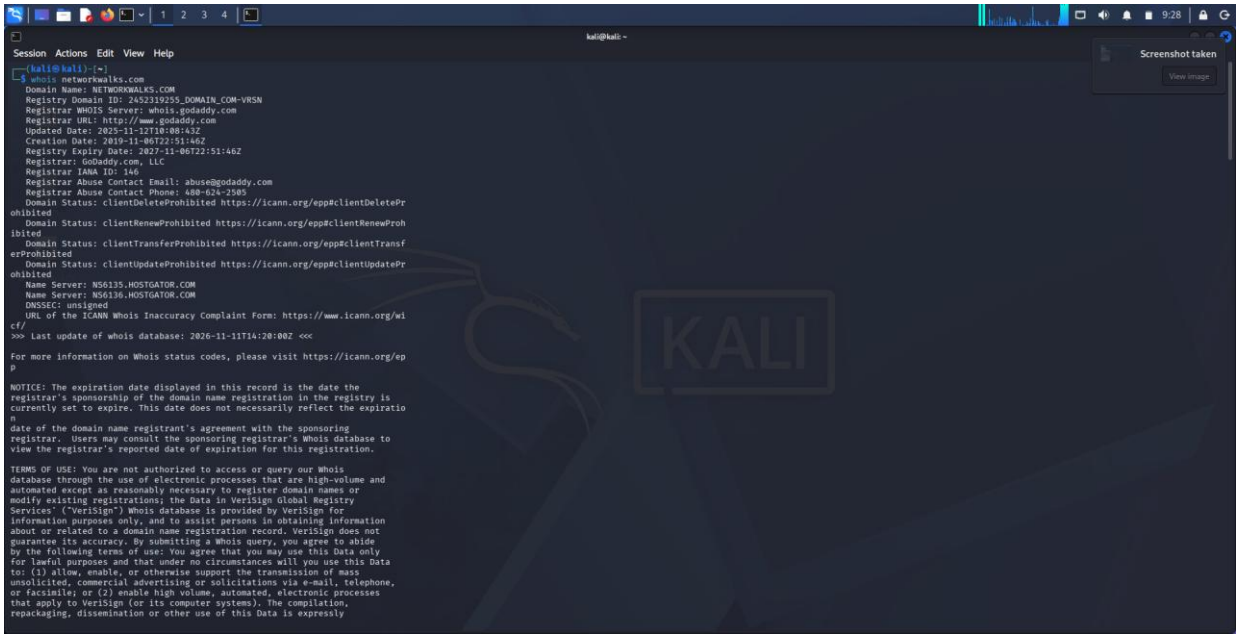


FIGURE 02 | WhatWeb - website technologies

WhatWeb output showing the web server, IP address and technologies reported for the website.



APPENDIX A / SCREENSHOTS

Evidence register (continued)

Continue inserting the evidence images below. Use the original screenshots from your practical session and do not recreate terminal output.

FIGURE 03 | nslookup - DNS resolution

Terminal output showing networkwalks.com resolving to 192.232.216.135



FIGURE 04 | curl - HTTP response headers

curl -I output showing the HTTP status and response headers



APPENDIX A / SCREENSHOTS

Evidence register (continued)

Continue inserting the evidence images below. Use the original screenshots from your practical session and do not recreate terminal output.

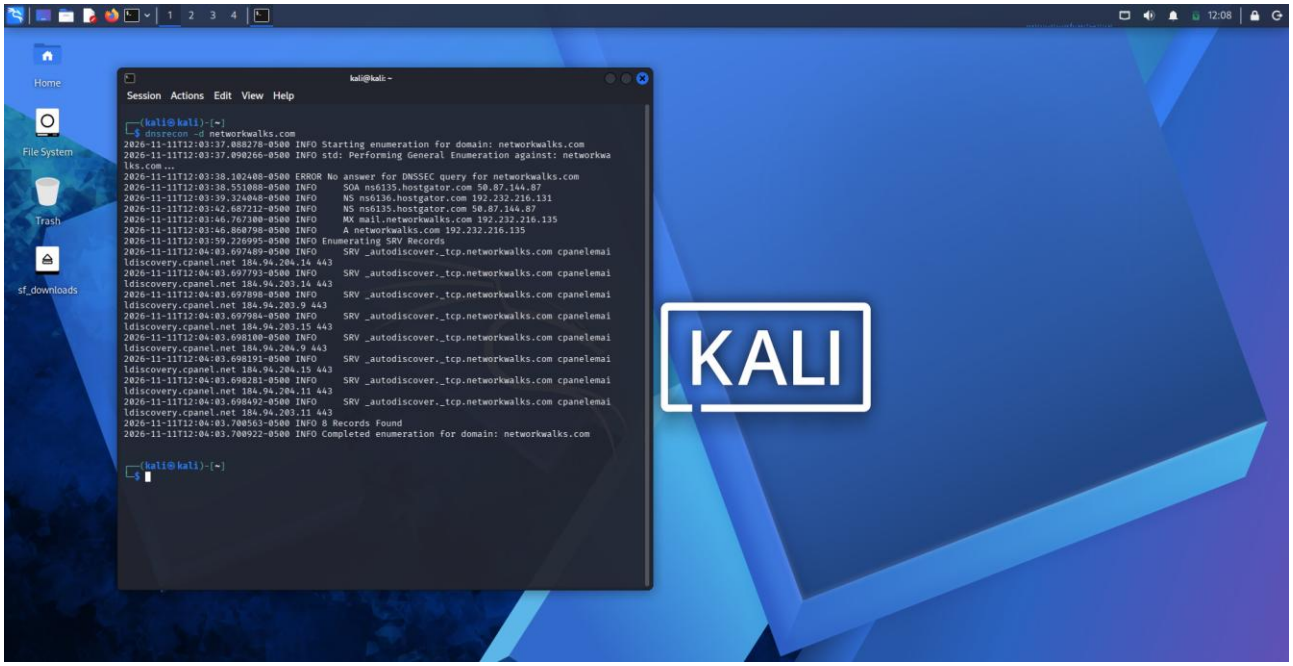
FIGURE 05 | Wafw00f - WAF identification

Wafw00f output identifying ModSecurity (SpiderLabs).



FIGURE 06 | DNSRecon - DNS records

DNSRecon output showing SOA, MX, TXT and SRV information.



APPENDIX A / SCREENSHOTS

Evidence register (continued)

Continue inserting the evidence images below. Use the original screenshots from your practical session and do not recreate terminal output.

FIGURE 07 | Zenmap - Ping Scan results

Zenmap result screen showing the 192.168.8.0/24 Ping Scan and responding hosts.

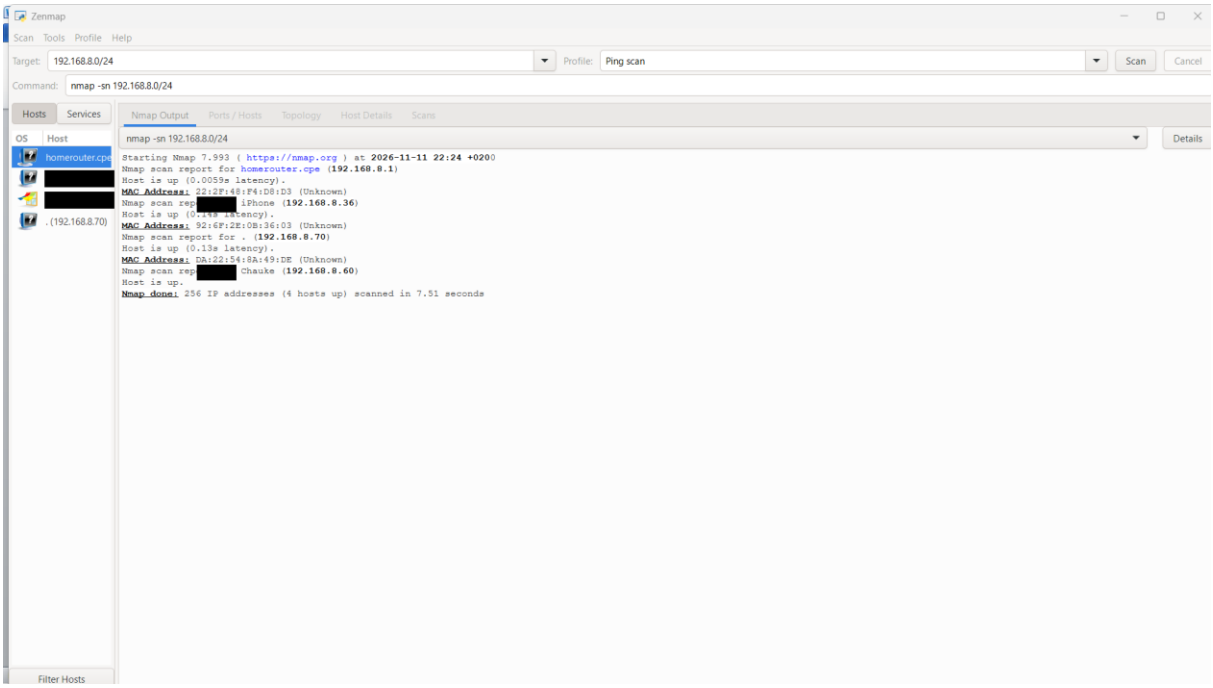


FIGURE 08 | Zenmap - network topology

Topology view with the legend enabled, or a clear screenshot of the saved topology PDF.

